

LAST NAME: _____

Practice Quiz 04D — MATH 2055 — Fall 2026

Practice for Quiz 04

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let $A = \{0\}$ and $B = \{1\}$.
 - (a) List all the elements of $\mathcal{P}(A) \cup \mathcal{P}(B)$ and of $\mathcal{P}(A \cup B)$.
 - (b) Exhibit a set that belongs to exactly one of the two families in part (a). Explain why it belongs to one and not to the other, and use this to decide whether $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ always holds.